IOPscience iopscience.iop.org

Home Search Collections Journals About Contact us My IOPscience

Dissociation of hydrogen molecules on metal filaments in H^- ion sources

This content has been downloaded from IOPscience. Please scroll down to see the full text.

1994 Plasma Sources Sci. Technol. 3 465

(http://iopscience.iop.org/0963-0252/3/4/003)

View the table of contents for this issue, or go to the journal homepage for more

Download details:

IP Address: 128.248.155.225
This content was downloaded on 03/10/2015 at 12:16

Please note that terms and conditions apply.

Plasma Source Sci. Technol. 3 (1994) 465–472. Printed in the UK

Dissociation of hydrogen molecules on metal filaments in H^- ion sources

A I Livshits†, F El Balghiti and M Bacal

Laboratoire de Physique des Milieux Ionisés, Laboratoire du CNRS, Ecole Polytechnique, 91128 Palaiseau, France

Received 21 January 1994, in final form 28 June 1994

Abstract. Atomic hydrogen causes significant deterioration of H^- ion source characteristics through vibration–translation relaxation of vibrationally excited H_2 molecules. We propose a method for atomic hydrogen density determination based on its selective absorption by a metal (Ta in our case) covered with a monoatomic non-metal film. The temperature-dependence of the dissociation rate by the Ta filament and the contribution of dissociation on Ta filaments to the atomic density in the H^- ion source are measured and reported. This contribution is noticeable but not dominant when the energy injected into the discharge is large enough. It is shown that the filament surface of W, Ta, Nb or W–Th behaves as a clean one with respect to H_2 dissociation under the actual conditions of H^- ion sources. According to a phenomenological theory, filaments of W, Ta and Nb dissociate H_2 independently of their temperature and of the gas pressure with probability about 0.2–0.3 per single impact, while a W–Th filament dissociates H_2 with much lower probability, which depends on pressure and temperature.

1. Introduction

The main production reaction of negative hydrogen ions, H^-, in volume sources is dissociative attachment of electrons to hydrogen molecules in highly vibrationally excited states [1]:

$$e + H_2(v'' \geqslant 5) \rightarrow H^- + H. \qquad (1)$$

One of the principal quenching processes of vibrationally excited H_2 molecules ($v'' \geqslant 5$) is vibration–translation transfer in collisions with hydrogen atoms. The reduction of the atomic hydrogen fraction is therefore extremely beneficial for H^- ion volume production [1].

The density of atomic hydrogen in the plasma of negative ion sources is determined by several production mechanisms [1] and by a single destruction process, recombination of atoms at walls. One of the atom formation mechanisms is dissociation of hydrogen molecules on the hot surface of the electron-emitting filaments [2].

In the meantime, deposition of filament material on the source wall is responsible for the recombination efficiency of this wall. We will analyse in this paper only the first of the two indicated aspects of the influence of the filaments upon the atomic density in H^- ion sources.

Dissociation of hydrogen molecules on hot metal surfaces has been analysed by several authors since Langmuir [3]. The most detailed analysis is given in the work of Brennan [4, 5], where the kinetic mechanisms of the process are discussed on a phenomenological level, and experimental data for W, Pt and Au are presented. The role of oxygen and carbon impurities on the surface is also discussed. Only recently have data on the dissociation of hydrogen on Ta and Nb been reported [6].

† Permanent address: Bonch-Bruyevich Institute for Telecommunications, Sankt Petersburg, Russia.

2. The principal equations

We will obtain here simple equations, which will lead to quantitative estimates and indicate the physical origin of the studied phenomenon. Let us consider the reaction

$$H_2 + Me \rightarrow 2H + Me. \qquad (2)$$

In order to determine the dissociation probability K_S of the H_2 molecule in one collision with the surface let us examine the equilibrium. The gas is in a vessel (a thermostat), the wall of which is made of the material of interest at temperature T, sufficient to bring about noticeable [7] dissociation. At equilibrium for reaction (2) the following relation applies:

$$\frac{P_H^2}{P_{H_2}} = K(T) \qquad (3)$$

where P_H and P_{H_2} are the partial pressures of atoms and molecules and K is the equilibrium constant. If the pressure is expressed in Torr, K can be found in the

0963-0252/94/040465 + 08$19.50 © 1994 IOP Publishing Ltd

range of temperature and pressure of interest from the following [8]:

$$K(T) = 760T^{1.72} \exp\left(-\frac{51900}{T}\right). \tag{4}$$

Let us define the degree of dissociation as follows:

$$\eta = \frac{0.5P_H}{0.5P_H + P_{H_2}}. \tag{5}$$

At equilibrium, η is

$$\eta_e = \left(\frac{K(T)}{K(T) + 4(P_H + P_{H_2})}\right)^{1/2} = \left(\frac{K(T)}{K(T) + 4P}\right)^{1/2} \tag{6}$$

where P is the total pressure.

At equilibrium the dissociation and recombination rates are equal:

$$K_S Z_{H_2} P_{H_2} = \tfrac{1}{2}\gamma_S Z_H P_H. \tag{7}$$

Here Z_{H_2} and Z_H are the gas-kinetic coefficients ($Z_H = \sqrt{2}Z_{H_2}$) and γ_S is the probability of H atoms recombination at one collision with the surface.

From equations (3) and (7) it follows that

$$K_S = \frac{1}{\sqrt{2}}\gamma_S\left(\frac{K}{P_{H_2}}\right)^{1/2}. \tag{8}$$

Further, reaction (2) includes at least two steps [5]: dissociative adsorption of an H_2 molecule:

$$\tfrac{1}{2}H_2 + Me \rightleftarrows HMe \tag{9}$$

and desorption of the H atom from the surface:

$$HMe \rightarrow H + Me. \tag{10}$$

Let us examine the cases when reaction (2) is limited either by the process described by equation (9) or by the process described by equation (10).

2.1. Desorption of H atoms, equation (10), is the limiting process

In this case we have the following.

(i) According to the rules of formal kinetics, we have to consider that there is equilibrium in the preceding step, equation (9), that is, the fluxes of adsorbed and desorbed molecules are almost equal.

(ii) In turn this means that the adsorbed atoms leave mainly in molecular form and only a small fraction are desorbed as H atoms.

(iii) From (ii) it follows that, if a free H atom has a collision with the surface and is adsorbed, then it is almost certainly desorbed in an H_2 molecule, that is, it recombines. Thus, in this case

$$\gamma_S = \alpha_H \tag{11}$$

where α_H is the probability of absorption of an H atom per collision with the surface.

(iv) From (ii) and (iii), it follows that the degree of dissociation $\eta_e \ll 1$, and consequently P_{H_2} does not depend on T, and $P_{H_2} \approx P$.

As a result, substituting $P_{H_2} = P$ in (8) we obtain

$$K_S = \frac{1}{\sqrt{2}}\gamma_S\left(\frac{K(T)}{P}\right)^{1/2}. \tag{12}$$

Note that here γ_S does not depend on P. This follows from the fact that, according to equation (11), $\gamma_S = \alpha_H$ and α_H does not depend on the surface coverage θ_H, nor on P, if $\theta_H \ll 1$. It is easy to show that $\theta_H \ll 1$ in that range of T and P in which dissociation actually occurs.

Thus, according to equation (12), $K_S \simeq P^{-1/2}$. The physical meaning of the increase of K_S when the pressure P is lowered is that, at lower P, the surface coverage with hydrogen, θ_H, is reduced. In turn this lowers the probability of an encounter of adsorbed atoms and of their desorption in a molecule, and increases the probability that adsorbed hydrogen will desorb as atoms.

Let us note that, in the discussed case, when $\eta \ll 1$ (iv), the equilibrium degree of dissociation η_e is, according to equation (6)

$$\eta_e = 0.5\left(\frac{K}{P}\right)^{1/2}. \tag{13}$$

Taking this into account, we have

$$K_S = \sqrt{2}\gamma_S\eta_e. \tag{14}$$

That is, the probability of dissociation K_S differs from the thermodynamically determined one for the given T and P value η_e only by the kinetic coefficient $\sqrt{2}\gamma_S$, which has a clear physical meaning.

2.2. The adsorption step, equation (9), is the limiting process

This means that, in contrast to the case in section 2.1, almost all the adsorbed hydrogen escapes as H atoms and so

$$K_S = \alpha_{H_2} \tag{15}$$

where α_{H_2} is the probability of dissociative adsorption per collision of the H_2 molecule with the surface (the sticking coefficient).

Equations (12) and (15) can be written, taking into account equation (4) as follows:

$$K_S = \min\begin{cases}\dfrac{1}{\sqrt{2}}\gamma_S\left(\dfrac{K}{P}\right)^{1/2} = 19.5\gamma_S P^{-1/2}T^{0.86}\exp\left(-\dfrac{25950}{T}\right)\\ \alpha_{H_2}\end{cases} \tag{16}$$

and so the dissociation probability is determined by the values of P and T and by the kinetic coefficients γ_S and α_{H_2}, which are independent of P (and often also of T) and are related to the particular surface.

3. The clean surface case

This case is important, when the surface of a thermionic emitter is involved. Owing to the high temperature (and in a plasma also due to ion sputtering) this surface can be very clean. The recombination coefficient γ_S on the clean surface of transition metals, including the high melting point metals, is independent of T and is close to unity [9, 10].

Relative to the sticking coefficient of molecules, α_{H_2} for transition metals (including the refractory ones) is usually independent of T and is in the range 0.1–0.3 [15]. In the dissociation experiments α_{H_2} was found to be 0.3 for W [5], 0.24 for Pt [5] and 0.2 for Nb and Ta [6]. Thus, in the case of a clean surface K_S increases when T is enhanced and when P is reduced (upper line of equation (16)), until saturation occurs ($K_S = \alpha_{H_2} =$ constant (P, T)). Taking $\gamma_S = 1$ and $\alpha_{H_2} = 0.25$, saturation occurs according to equation (16) for

$$P^{-1/2}T^{0.86}\exp\left(-\frac{25950}{T}\right) \sim 0.01. \qquad (17)$$

Equations (16) and (17) are plotted in figure 1 by isobars for a few pressures that are usual for negative hydrogen ion sources. Note that, for pressures lower than 3×10^{-2} Torr, saturation of the dependence of K_S on T occurs already for $T \geqslant 2100$ K and we get

$$K_S = \alpha_{H_2} = \text{constant}\ (T) \approx 0.2\text{–}0.3. \qquad (18)$$

Since much higher temperature is required (figure 1) to obtain thermionic emission from W, Ta and Nb, filaments made of these metals dissociate hydrogen with the probability $K_S = 0.2$–0.3, independent of T and P (on condition that their surface is not covered by a non-metallic film).

On the other hand, a filament of thoriated tungsten emits several A cm^{-2} at $T \simeq 1750$ K. At this T, the probability of dissociation is considerably lower than the maximum value, and is dependent on P and T, and for given P and T it can be found from figure 1, or calculated from the upper line of equation (16).

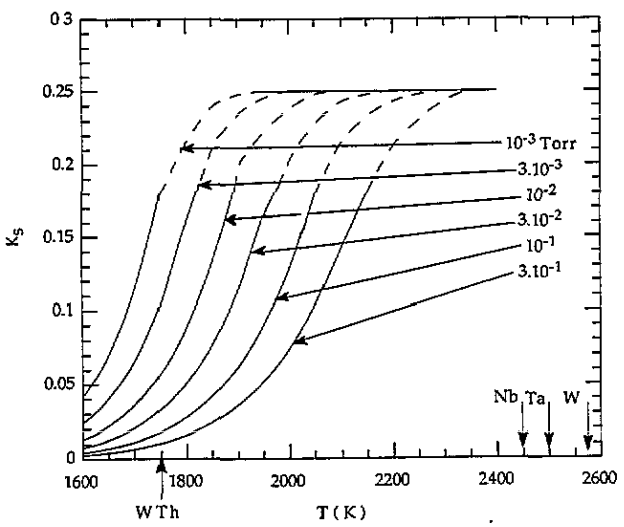


Figure 1. Dependence on surface temperature of the dissociation probability calculated from equation (16), at $\gamma_S = 1$, $\alpha_{H_2} = 0.25$. The arrows indicate the surface temperatures at which W, Ta, Nb and W–Th produce an emission current of 2 A cm^{-2} [30]. The broken lines correspond to the transition between the dissociation regimes described by the upper and lower lines of equation (16).

4. The role of non-metallic contaminants

Even monoatomic non-metallic films can cancel the catalytic action of a metal surface. Therefore it can be questioned whether the filament surface will act as a clean one with respect to hydrogen dissociation under the conditions of plasma ion sources.

The usual impurities on surfaces are carbon, oxygen and sulphur. Sulphur does not stay on the surface at the usual temperatures of thermionically emitting filaments, and its compounds (such as H_2S) are usually absent from the residual gas of vacuum stands. Therefore, once cleaned of S, the surface is no longer contaminated by it. However, films containing C and O can be dynamically maintained on the surface due to the presence in the residual gas of CO, H_2O, CO_2, O_2, CH_4 and often of heavier organic molecules. Let us examine separately contamination by C and by O.

Contamination of thermionic filaments by carbon is usual. W, Ta and Nb at their operating temperature can dissolve considerable quantities of carbon (W at 2580 K up to 0.2 at%, Ta at 2470 K up to 2.5 at% and Nb at 2450 K up to 2 at% [16]). At higher concentrations of carbon, a carbide phase known as the temperature-resistant phase separates out. However, as shown by direct experiments, neither the solution of C, nor the carbide phase, have an important effect upon the catalytic properties of the W, Ta and Nb surfaces.

Hörz *et al* [17] have shown that the presence of a few atomic percent of carbon in Ta and Nb does not change the dissociation rate of methane and other hydrocarbons on their surfaces. The same observation was made for hydrogen dissociation on surfaces of tantalum [18] and tungsten [4]. Only when new doses of carbon are introduced after full carbidization and graphite is formed on the surface of the carbide do the surface properties drastically change in such a way that catalytic processes tend to be slowed [19]. A graphite film is unacceptable on the surface of a thermionic emitter, because of the considerable increase in work function. Thus, in practice carbon is not a 'poison' in hydrogen dissociation on W, Ta and Nb. The considerations of section 3 relative to the clean surface also apply, to a first approximation, in the presence of carbon.

In contrast to the case of carbon, oxygen radically cancels the catalytic action of W, Ta and Nb. This was shown by direct experiments on hydrogen dissociation on W [4, 5] and Ta [18], and also in experiments on

dissociation of methane and other hydrocarbons on Ta and Nb [17]. In the case of hydrogen dissociation on W, the poisoning effect of oxygen is not important, because the tungsten oxides are reduced in hydrogen at temperatures as low as 1200 K [5, 16]. However, in the case of Ta and Nb, hydrogen does not chemically clean oxygen from the surface [16, 17] and oxygen is removed by such processes as thermal evaporation of oxides, reactions with hydrocarbons like

$$Me_xO_y + C_nH_m \rightarrow Me + CO\uparrow + H_2\uparrow$$

and sputtering. The arrival of oxygen on the surface can be due to O_2 and H_2O in the gaseous phase. As a result a stationary coverage θ_0 by oxygen of the surface is established, and, if $\theta_0 > 0.5$, this leads to reduction by orders of magnitude of the probability of catalytic reactions [17], including dissociation of H_2 molecules [18]. In the absence of leaks, the pressure of H_2O usually considerably exceeds the pressure of O_2 [20].

The minimum water vapour pressure P_{H_2O}, for which $\theta_0 \gtrsim 0.5$ is maintained on filaments of Ta and Nb, can be evaluated by assuming that removal of oxygen occurs only by evaporation of oxides.

As a result of H_2O action and evaporation of oxides, a steady-state concentration of dissolved oxygen is established within the volume of Ta [16]:

$$C_o = 1.35 \times 10^{-2} P_{H_2O} \exp\left(\frac{27320}{T}\right) \qquad (19)$$

where C_o is in atomic percent and P_{H_2O} in Torr. On the other hand, the Ta surface coverage is determined by the concentration of dissolved oxygen C_o [17]:

$$\frac{\theta_0}{1-\theta_0} = 0.4 \exp\left(\frac{4940}{T}\right) C_o \qquad (20)$$

From (18) and (19) we find the value of P_{H_2O} responsible for the coverage θ_0:

$$P_{H_2O} = 185 \exp\left(-\frac{32260}{T}\right) \frac{\theta_0}{1-\theta_0}. \qquad (21)$$

With $\theta_0/(1-\theta_0) = 1$ and $T = 2470$ K (the temperature at which the density of thermionic emission is $2\,A\,cm^{-2}$), we obtain $P_{H_2O} = 4 \times 10^{-4}$ Torr. This is a very high partial pressure of water vapour. For example, in our hybrid H^- ion source, which is continuously pumped by a turbomolecular pump ($50\,l\,s^{-1}$), this pressure, measured using a quadrupole mass spectrometer, is at least 100 times less [21]. Thus, under actual conditions the surface of the tantalum filament is practically free of oxygen [22]. This result can also be shown for Nb.

In conclusion, the usual impurities (C and O) should not influence considerably the dissociation of hydrogen by filaments made of W, W–Th, Ta and Nb. Therefore the data for dissociation on a clean surface of these materials can be used.

5. The contribution of the filament surface to production of atomic hydrogen in the hybrid H^- ion source

The density of hydrogen atoms in a H^- ion source has been determined using a laser diagnostic [2] and translational spectroscopy [23]. Here we propose an absorptional spectroscopy [23]. Here we propose an absorption method for determining P_H and present the results of measurements in a hybrid H^- source [24] with tantalum filaments.

5.1. An absorption method for determining the hydrogen atom density

This method is based on the property of a metal surface covered with a monoatomic film of a non-metallic impurity to selectively absorb atomic hydrogen [14]. A monoatomic film of an impurity can reduce the absorption probability of hydrogen molecules, α_{H_2}, by orders of magnitude with respect to the case of a clean surface; as a result, molecular hydrogen practically stops being absorbed. In the meantime this film does not prevent absorption of atomic hydrogen: the thermal hydrogen atoms continue to be absorbed with probability $\alpha_H = 0.1-1$, independently of the metal temperature. The mentioned film also increases the ability of the metal to retain hydrogen. Correspondingly, during interaction of the metal with atomic hydrogen, the established density of hydrogen atoms dissolved in the metal increases by orders of magnitude, compared with the case of a clean surface. Monoatomic non-metallic films are chemically stable in atomic hydrogen. The monoatomic state of the film, at not very low temperature (about 700 K and higher) is maintained automatically due to exchange of impurity atoms between the metal volume and its surface (the phenomenon of surface segregation of impurities [17, 19, 25].

The experimental set-up is shown in figure 2. The component that selectively absorbs atomic hydrogen is a directly heated ribbon of tantalum 50 μm thick and of surface area 420 cm^2. In the initial state the H_2 pressure is established, the dissociation source (for example the filament) is operated and the absorbing ribbon has a high enough temperature T_r (such as 1300 K), at which its absorption capacity is low. Therefore the steady-state density of atomic hydrogen is rapidly (during a few seconds) established in it and the absorption stops. Then we rapidly lower the temperature of the ribbon (for example to 700 K) and it starts absorbing atomic hydrogen. The value of the initial absorption rate (independent of T_r) exceeds up to 20 times the speed of pumping of the discharge chamber by the turbomolecular pump (about $25\,l\,s^{-1}$ for H_2). In our set-up the hydrogen pressure in the discharge chamber is maintained constant using a gas flow control device (figure 2). Therefore the absorption leads to a considerable increase in H_2 flow into the system (by up to a factor of 20). The increase in flow ΔJ (figure 3) is directly recorded by

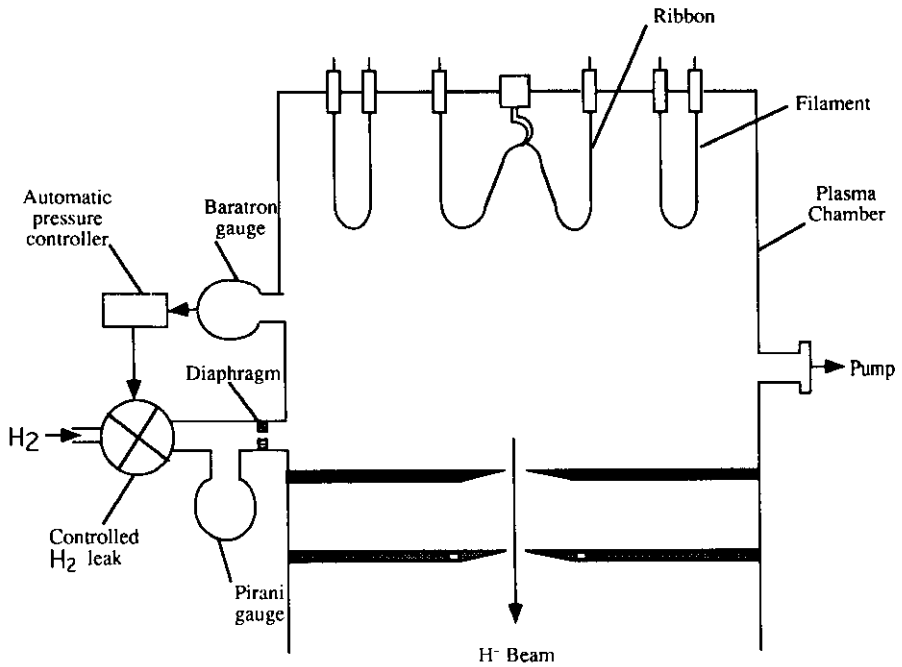


Figure 2. Schematic representation of the hybrid H⁻ source and of the set-up for measuring the atomic density.

observing the pressure increase with a Pirani gauge (see figure 2) in the gas supply system, separated from the discharge chamber by a diaphragm of sufficiently low conductance (opening 3 mm diameter, 8 mm long). The inertia of the measuring device is sufficiently low and does not affect the result. The hydrogen flow, corresponding to some indication of the Pirani gauge (figure 2), is determined using an absolute membrane pressure gauge (Baratron) (figure 2) at a known (specially determined) speed of pumping of hydrogen by the turbomolecular pump (approximately 25 $l\,s^{-1}$) [26].

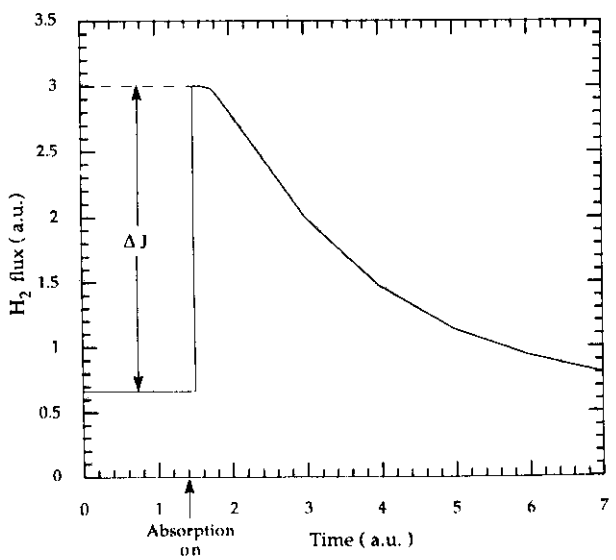


Figure 3. A typical curve of hydrogen flow enhancement in the discharge chamber while the tantalum ribbon cools in atomic hydrogen ('absorption on'). The characteristic sorption time depends strongly on the ribbon temperature and, in our experiments, it was of several minutes duration.

The flow absorbed by the tantalum ribbon (figure 2) in the initial absorption phase, when the reverse flow from the ribbon is still absent, is related to atomic hydrogen pressure by

$$\Delta J = 0.5 P_H Z_H \alpha_H B_r \tag{22}$$

where α_H is the probability of absorption of atomic hydrogen per collision with the absorbing ribbon area and B_r is the area of the ribbon surface. In the general case, when the temperatures of the atoms, T_H, and of the molecules, T_{H_2}, are different, the degree of gas dissociation is determined as follows (compare with equation (5)):

$$\eta = \frac{0.5 P_H/(KT_H)}{0.5 P_H/(KT_H) + P_{H_2}/(KT_{H_2})}. \tag{23}$$

With $\eta \ll 1$, we obtain from (22) and (23)

$$\eta = \frac{\Delta J}{P_{H_2} Z_H \alpha_H B_r} \frac{T_{H_2}}{T_H} \tag{24}$$

Furthermore

$$P_{H_2} = P^0_{H_2} \left(\frac{T_{H_2}}{293}\right)^{1/2} \tag{25}$$

where $P^0_{H_2}$ is the automatically maintained pressure in the Baratron gauge (figure 2). Equation (25) is a consequence of the Knudsen regime of gas exchange between the gauge and the discharge chamber at pressures below 10^{-2} Torr. Taking into account equation (25) and substituting

$$Z_H = Z_H(293) \left(\frac{293}{T_H}\right)^{1/2}$$

(where $Z_H(293)$ is the gas-kinetic coefficient for atoms at room temperature), we obtain from equation (24)

$$\eta = \frac{1}{P^0_{H_2} Z_H(293) B_r} \frac{1}{\alpha_H} \left(\frac{T_{H_2}}{T_H}\right)^{1/2} \Delta J. \qquad (26)$$

Assuming that α_H does not depend on T_H [27], we obtain for $T_H = T_{H_2}$

$$\eta = A \Delta J \qquad (27)$$

where A is a coefficient, independent of gas temperature, given by

$$A = \frac{1}{P^0_{H_2} Z_H(293) \alpha_H B_r}. \qquad (28)$$

If α_H is known, then equations (27) and (28) also lead to the absolute value of η.

5.2. The dissociation probability on the tantalum filament surface in the absence of plasma

The atomic hydrogen formed on the filament recombines essentially on the cold walls. This leads to the balance

$$2P_{H_2} Z_{H_2} K_S B_f = P_H Z_H \sum_i \gamma_{si} B_i. \qquad (29)$$

Here B_f is the area of the filament surface, γ_{si} are the recombination coefficients on various surfaces i (including the wall) and B_i are the area of each of these surfaces. If we would take further into account that Z_H and Z_{H_2} are determined by the wall temperature, and also that α_H and γ_{si} are independent of the hydrogen pressure [9, 11–14] and that $P_H \ll P_{H_2}$, then there follows from equations (22) and (29) the proportionality

$$K_S = A' \Delta J$$

where A' is a coefficient independent of cathode temperature and gas pressure.

For example, it is possible to find the temperature-dependence of K_S by measuring ΔJ for various cathode temperatures. The results of such measurements are shown in figure 4. In this experiment, before each measurement of K_S at relatively low temperature ($T < 2500$ K), the filament is maintained at 2500 K for about 10 s, after which it takes about 1 min to establish the necessary temperature and effect the measurement.

From comparison of experimental data with calculations, it can be noted that, under the usual physical and chemical conditions [28], tantalum dissociates hydrogen in a very similar way to that predicted for a clean surface at the given pressure: no dependence of K_S on T is observed at high temperatures and transition to saturation with T occurs at $T \approx 2000$ K. If it is assumed, as in figure 4, that $K_S = 0.2$ at saturation for tantalum [6], then the experimental values for K_S at low T (1500–1700 K) and $P = 3 \times 10^{-3}$ Torr, coincide with the calculated ones (the upper line in equation (16)) at $\gamma_S \approx 0.5$. This recombination coefficient value is about half that found for a clean tantalum surface [9]. This can be

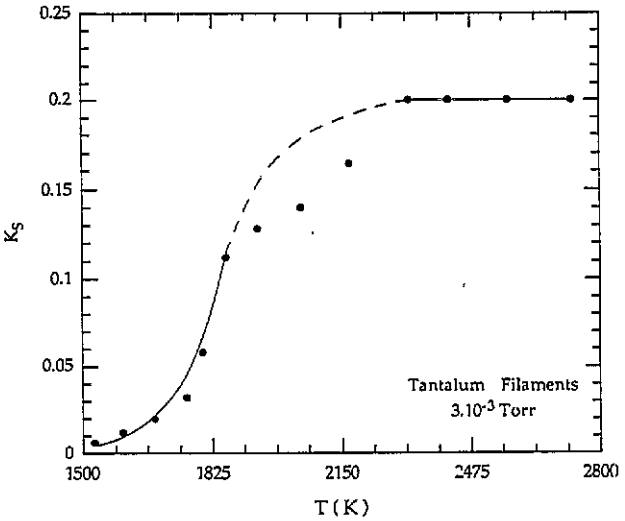


Figure 4. Dissociation probability on the tantalum surface. Full circles are results of the experiment with tantalum filaments in the hybrid H^- ion source. The absolute scale of K_S is defined assuming that, at saturation, $K_S = 0.2$, as was found for tantalum [6]. The full line shows the calculation using equation (16) for a clean surface with $\gamma_S = 0.5$ and $\alpha_{H_2} = 0.2$.

easily explained by the fact that the tantalum surface may not be absolutely clean at a relatively low (<1700 K) temperature [22] and under the vacuum conditions of the present H^- source.

5.3. The role of dissociation at the cathode surface in the discharge

In figure 5 is shown the observed change in gas flow into the system, ΔJ (see figure 3) determined by the absorption of 'hot' hydrogen by a tantalum ribbon as a function of the power (the current–voltage product), applied to the discharge. Power to the tantalum filament

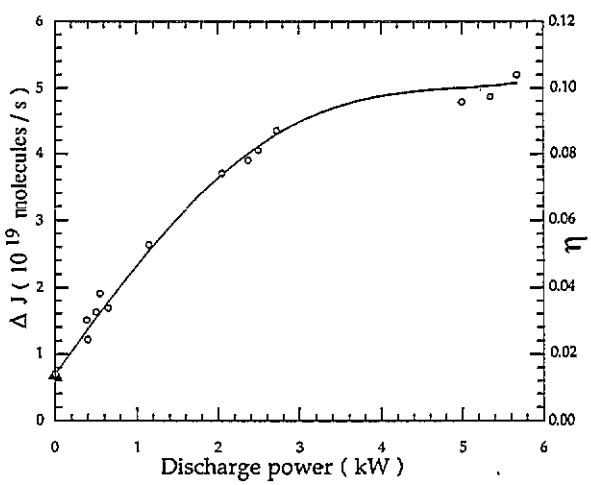


Figure 5. The observed change of gas flow ΔJ (figure 3), due to atomic hydrogen absorption, and the degree of gas dissociation η versus the discharge power. When the discharge power is zero (discharge off), dissociation occurs only on the hot filament. The absolute scale of η is defined with the assumption that $\alpha_H = 0.2$ and $T_H = T_{H_2}$.

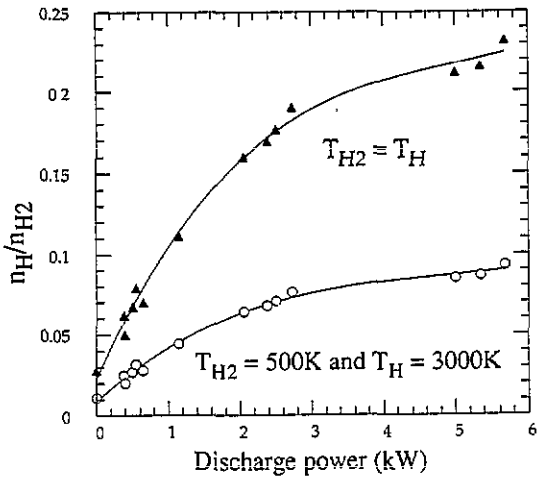


Figure 6. Relative atomic hydrogen density n_H/n_{H_2} versus discharge power, with $T_H = T_{H_2}$ and with $T_{H_2} = 500$ K and $T_H = 3000$ K.

is kept switched on also when the discharge is switched off. In this case the cause of absorption is dissociation on the filament surface. The experiment was effected at pressure $P \approx 3 \times 10^{-3}$ Torr [29].

First of all, we have to understand whether the results of this experiment should be ascribed to atomic hydrogen, despite the fact that ions present in the plasma may also be absorbed by the tantalum ribbon. The ratio of the absorbed fluxes of atoms and ions is

$$\frac{\eta\sqrt{T_H}\alpha_H}{\varpi\sqrt{T_+}\alpha_+}$$

where T_H and T_+ are the atomic and ionic temperatures, η and ϖ are the degrees of gas dissociation and ionization respectively and α_H and α_+ are the probabilities of absorption of an atom and of an ion by tantalum per collision respectively.

At pressure 3×10^{-3} Torr the ratio η/ϖ is of the order of 100, $(T_H/T_+)^{1/2} \approx 0.5$–1. Even if we assume that $\alpha_H/\alpha_+ = 0.1$ (usually $\alpha_H \geqslant 0.1$ [14]), we must conclude that the flux of absorbed 'hot' particles from the plasma is due to atomic hydrogen.

If we assume that $(T_{H_2}/T_H)^{1/2}$ does not change significantly with change in energy injected into the plasma, then we obtain according to equation (26) $\eta \approx \Delta J$, which allows us to represent in figure 5 the degree of dissociation η. With the assumptions that $\alpha_H \approx 0.2$ [6] and $T_H \approx T_{H_2}$, we obtain from equations (27) and (28) the absolute values of η (figure 5).

In figure 6 is shown the relative atomic hydrogen density n_H/n_{H_2}, deduced from (23), versus the power applied to the discharge, with both $T_H \approx T_{H_2}$ and realistic values of $T_H = 3000$ K and $T_{H_2} = 500$ K. The use of realistic values instead of assuming equal temperatures leads to reduction of the maximum relative atomic density from 0.22 to 0.09. Comparing the n_H/n_{H_2} values (figure 6) in the discharge with the n_H/n_{H_2} values without discharge (but with the filaments switched on), we conclude that, at large injected energy into the discharge, the contribution of dissociation on the filaments to the atomic concentration is not dominant, although it remains noticeable. The maximum ratio n_H/n_{H_2} measured in the large volume hybrid H^- source used in the present experiment (0.09) is larger than the maximum n_H/n_{H_2} measured by translational spectroscopy [23] in a smaller hybrid source. This confirms the theoretical scaling law [31], which suggests that the atomic density is increased with a larger chamber volume. The results of figure 6 and [23] indicate that the relative atomic density saturates at high power, instead of increasing linearly with discharge power as indicated by theoretical scaling. We assume that this is related to the change in wall recombination coefficient when the discharge power is varied.

6. Conclusion

(i) Under the actual physico-chemical conditions of a plasma device (the H^- source) the tantalum surface (and also, probably surfaces of Nb, W and W–Th) is clean of oxygen and behaves as a clean surface with respect to dissociation of H_2 molecules.

(ii) Ta, Nb and W cathodes at temperatures suitable for thermionic emission dissociate hydrogen independently of temperature and pressure with probability 0.2–0.3 per collision. The dissociation probability on a W–Th filament is considerably lower and depends on P and T.

(iii) For a discharge power of more than 1 kW in a volume of 0.1 m^3, dissociation on a filament surface of Ta (also Nb or W) is observable but not dominant.

(iv) The absorption method for detection of atomic hydrogen is very adequate for plasma investigations.

(v) Saturation of the relative atomic density at high discharge power was observed, confirming earlier observations [23].

Acknowledgments

The authors acknowledge financial support from the Direction des Etudes, Recherches et Techniques (France) and useful discussions with Dr C Courteille and Dr Z. Sledziewski.

References

[1] Bacal M and Skinner D A 1990 *Comments At. Mol. Phys.* **23** 283
[2] Eenshuistra P J, Heeren R M, Kleyn A W and Hopman H J 1989 *Phys. Rev.* A **40** 3613
[3] Langmuir I 1915 *J. Am. Chem. Soc.* **37** 417
[4] Brennan D and Fletcher P C 1959 *Proc. R. Soc.* A **250** 389
[5] Brennan D 1964 *Adv. Catalysis* **15** 1
[6] Livshits A, Notkin M, Samartsev A, Busnjuk A, Doroshin A and Pistunovich V 1992 *J. Nucl. Mater.* **196–198** 159

[7] A system with hot filaments, cold walls and a plasma is obviously a non-equilibrium one. However, under conditions of a small coverage of a hot surface, which is always the case, and also for the usually non-activated processes of atomic recombination and molecule sticking, the lack of equilibrium does not influence K_S. In particular, this follows from analysis of the relation obtained by Brennan [8], and also from comparison with experimental data.

[8] Brennan D, Hayword D O and Trapnell B M W 1960 *Proc. R. Soc.* A **256** 81

[9] Beitel G A 1969 Recombination of atomic hydrogen on metal in ultra high vacuum *PhD thesis* University of Wisconsin

[10] According to the data of other authors [11–13] γ_S for different metals is in the range 0.05–0.3. However, in these experiments the surface was not clean. Beitel [9] proved this, by finding γ_S in the mentioned range 0.05–0.2 for insufficiently cleaned surfaces, while on a clean surface $\gamma_S \approx 1$. This was also confirmed by work on superpermeation [14], in which H atoms are absorbed and then recombine on metals, the surface of which is covered by a monoatomic non-metallic film, with recombination probability 0.1–0.3, independently of the temperature.

[11] Wise H and Wood B J 1967 *Adv. At. Mol. Phys.* **3** 291

[12] Wood B J and Wise H 1961 *J. Phys. Chem.* **65** 1976

[13] Lavrenko V A 1973 *Recombination of Hydrogen Atoms on the Surfaces of Solids* (Kiev: Naucova Dumka)

[14] Livshits A, Notkin M and Samartsev A 1990 *J. Nucl. Mater.* **170** 79

[15] Roberts M W and McKee C S 1978 *Chemistry of the Metal–Gas Interface* (Oxford: Clarendon)

[16] Fromm E and Gebchardt E (ed) 1976 *Gase und Kohlenstoff in Metallen* (Berlin: Springer)

[17] Hörz G, Kanbach H and Vetter H 1980 *Mater. Sci. Eng.* **42** 145

[18] Livshits A, Notkin M and Busnjuk A 1991 private communication

[19] Rutkov E V, Tontegode A Ja 1982 *Sov. Phys.–Tech. Phys.* **52** 921

[20] This is especially characteristic for systems that are not heated, and also for work with hydrogen, particularly atomic hydrogen.

[21] In fact, pumping of water vapour (and also of oxygen) is not due to the pump, but to the tantalum filaments: with switching on of the filament, the water vapour pressure decreases by more than an order of magnitude.

[22] Let us note that this result is due to the very high filament temperature. For instance, at 1800 K, for an oxygen-free surface, according to equation (21) a pressure $P_{H_2O} \leqslant 3 \times 10^{-9}$ Torr is required.

[23] Bruneteau A M, Hollos G, Bacal M and Bretagne J 1990 *J. Appl. Phys.* **67** 7254

[24] Courteille C, Bruneteau J, Valckx F P G, Sledziewski Z and Bacal M 1993 *Rev. Sci. Instrum.* **64** 3265

[25] Benardt J 1983 *Absorption on Metal Surfaces: An Integrated Approach* (Amsterdam: Elsevier)

[26] This eliminates an inconvenience related to the nonlinearity of the Pirani gauge and the complex dependence of the conductance of diaphragm (figure 2) on pressure at the transition from the molecular to the viscous flow regime.

[27] Let us remind ourselves that absorption of atoms is an unactivated process [14] and therefore the dependence of α_H on T_H can be related only to some changes in the reflection coefficient of atoms incident at the surface.

[28] The plasma device is not heated; it has viton gaskets, and is continuously pumped by a turbomolecular pump with a pumping speed of $50\,l\,s^{-1}$ for air. At the tantalum filament temperature of 2400 K and a temperature of the absorbing tantalum ribbon of 850 K the pressures of H_2O, CO plus N_2, CO_2 and CH_4 are, respectively, 5×10^{-7}, 1×10^{-6}, 3×10^{-8} and 1.5×10^{-7} Torr. Heavy organic compounds are absent.

[29] In fact, this pressure was maintained by a pressure control system, activated by the Baratron pressure gauge (figure 2), connected to the discharge chamber by a tube. Since gas exchange between gauge and discharge chamber occurs in the pressure range 10^{-3}–10^{-2} Torr in the Knudsen regime, the pressure in the discharge chamber exceeds the pressure in the gauge by a factor of $(T_H/293)^{1/2}$ and consequently can increase by an order of magnitude when the power (and thus gas temperature) is enhanced.

[30] Dushman S and Lafferty J M 1966 *Scientific Foundations of Vacuum Techniques* 2nd edn (New York: Wiley)

[31] Skinner D A, Bruneteau A M, Berlemont P, Courteille C, Leroy R and Bacal M 1993 *Phys. Rev.* E **48** 2122